



Daniel (Tuan) PHAM

DATA SCIENTIST / AI ENGINEER /
RESEARCH ENGINEER

PH.D. IN COMPUTER SCIENCE

Valid Luxembourg Work Permit
Fast-Track Employer Transfer

CONTACT

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EDUCATION

- 2020 - 2023**
Ph.D. in Computer Science
University of Lorraine, Metz,
France
- 2014 - 2016**
**MSc. in Information
Technology**
University of Lorraine, Metz,
France
- 2008 - 2012**
**BS in MIS - (Hybrid: IT &
Economics)**
National Economics University,
Vietnam

LANGUAGES

- ✓ Vietnamese: Native
- ✓ English: Professional Working
- ✓ French: Elementary

PROFESSIONAL SUMMARY

- **Ph.D. in Computer Science** with **10+ years of experience** bridging theoretical research and enterprise deployment, transforming complex scientific and business challenges into scalable production AI systems.
- Proven track record engineering **high-throughput ML pipelines**, **geospatial analytics platforms**, and **agentic workflows**, delivering end-to-end solutions from mathematical modeling to full-stack cloud deployment.

WORK EXPERIENCE

- Senior Data Scientist** 10/2024 - Present
Databourg-Komunidad Global, Luxembourg
 - **Real-time Weather Forecasting (Rainvision):** Led development of multi-source weather forecasting models, processing data from **60,000+ satellite and IoT sensors** worldwide; boosted prediction accuracy by **5%** by migrating legacy ML to scalable Deep Learning architectures.
 - **Full-Stack GIS & Pipeline Architecture:** Re-architected data pipelines using **Airflow, RabbitMQ, & Docker** to automate workflows and enhance system throughput by **20-40%**; delivered commercial-grade interactive map dashboards and secure APIs using **FastAPI, Django, MongoDB, and MapboxGL** for public agencies across France, the Philippines, Indonesia, and Brazil.
 - **Climate Risk Scoring Platform:** Engineered specialized analytics platform evaluating physical climate risks and their financial impacts on operations, processing **multi-terabyte netCDF environmental datasets** for global sustainability regulations compliance.
- Postdoctoral / R&D engineer** 12/2023 - 06/2024
LCOMP - Univ. of Lorraine, Metz, France
 - **Technical Leadership (DCASOLVER):** Served as Tech-Lead for the **DCASOLVER** project, architecting and launching a commercial web platform; developed mathematical optimization and machine learning algorithms for **GIS logistics, spatial routing, and multi-hub warehousing**.
- Ph.D. Candidate / R&D engineer** 10/2020 - 11/2023
University of Lorraine, Metz, France
 - **Ph.D. Research:** Authored doctoral thesis on machine learning for finance and healthcare, designing novel DCA optimization algorithms and fine-tuning transformer models (BERT/RoBERTa) for clinical text classification.
 - **Defense R&D Partnerships:** Executed 3 joint research projects with naval contractor NAVAL Group, applying Deep Learning, Computer Vision, and autonomous robotics.
- University Lecturer & Researcher** 08/2012 - 10/2020
National Economics University, Hanoi, Vietnam
 - Taught core courses: Fundamental Computer Science, Applied Informatics, and MIS.
 - Conducted academic research in AI; maintained concurrent industry engineering roles (VinTech, NAL, Chappiebot) to deploy R&D models into commercial applications.
- GIS Data Engineer / Web Developer** (Concurrent / Hybrid) 08/2019 - 12/2019
VinTech Group, Hanoi, Vietnam
 - Held concurrent industry role alongside university post: Implemented an AI-based Logistics web system with Django, Bootstrap, GoogleMap APIs for Vinmart's retail network (2,000+ stores across Vietnam).
 - Collected data, built ETL processing pipelines, designed MySQL databases, and deployed services on AWS cloud.

SKILLS

CORE AI & DS

Machine Learning

Deep Learning

LLMs & NLP

Advanced RAG

Data Processing

GEOSPATIAL

GDAL/Rasterio

netCDF/HDF5

MapboxGL/Leaflet

QGIS

Spatial Analysis

DATABASE & INFRA

PostgreSQL/pgvector

MongoDB

Docker

Airflow/RabbitMQ

MLflow

WEB & CLOUD

Django/FastAPI

React

AWS/Azure/GCP

LANGUAGES

Python (OOP)

SQL

C/C++

PRACTICES & TOOLS

PyTorch/TensorFlow

Sklearn/Pandas

Mathematical Optimization

OpenCV

CI/CD Pipelines

Agile/Scrum

CERTIFICATIONS

 Generative AI with LLMs - AWS, 2024

 MLOps with Azure - LinkedIn, 2024

 MLOps Essentials - LinkedIn, 2024

 Intro to Large Language Models - LinkedIn, 2024

 Deep Learning Specialization - Coursera, 2018

 Data Science in Healthcare - Univ. Edinburgh, 2023

 Data Science Ethics - Univ. Michigan, 2023

HOBBIES

 Running / Football

 Reading / Learning

 Taking care of family

• **ML Engineer / Product Owner** (Concurrent / Hybrid) 12/2017 - 07/2019

 *NAL Jsc., Hanoi, Vietnam*

- Held concurrent industry engineering role alongside university post: Led an 8-person team conducting AI R&D.
- Delivered production AI-based solutions for external clients: Operator Chatbot, Iris recognition & tracking, Face lock verification, Dialogue Systems, and Hate Speech Detection.

• **NLP Engineer** (Concurrent / Hybrid)

10/2016 - 10/2017

 *Chappiebot Inc., Hanoi, Vietnam*

- Held concurrent industry role alongside university post: Developed core machine learning algorithms for an AI-based car search engine.
- Engineered Vietnamese NLP pipelines: Named Entity Recognition (NER), Topic Classification, Sentiment Analysis, and intent matching for Dialogue Systems.

• **ML Research Intern**

04/2016 - 10/2016

 *LITA - Univ. of Lorraine, Metz, France*

- Thesis: "Machine learning techniques for Autonomous Surface Vessels".
- Conducted research on long-term autonomy and data acquisition for environmental monitoring using Reinforcement learning, robotics control techniques on Unmanned Surface Vehicles (USV).

KEY PROJECTS

▶ **Climate Risk Scoring Platform (2025):** Analytics platform evaluating physical climate risks and financial impacts for global sustainability regulations compliance. *(ESG Data, netCDF, GDAL/Rasterio, Python, Statistics, MongoDB, React)* .

▶ **Real-time Weather Forecasting (Rainvision) (2024):** Multi-source forecasting models processing 60,000+ sensors; deployed interactive GIS dashboards across Philippines, Indonesia and Brazil. *(Deep Learning, Airflow, RabbitMQ, Docker, FastAPI, MapboxGL)* .

▶ **Legal & Regulatory GraphRAG Engine (2025 - Present):** High-precision knowledge retrieval engine for complex, hierarchical legal corpora; implemented **Advanced RAG with Hybrid Search (BM25 + Dense embeddings)** and **Cross-Encoder Reranking** to minimize hallucinations. Engineered a **GraphRAG** pipeline mapping legal cross-references and statutory hierarchies across multi-node vector indices. *(Advanced RAG, GraphRAG, Qdrant, pgvector, Hybrid Search, Reranking)* .

▶ **DCASOLVER Platform (2024):** Tech-Lead for AI platform providing mathematical optimization and machine learning solutions for large-scale transportation and warehousing systems. *(Python, Django, FastAPI, MySQL, Linux)* .

▶ **Robot navigation for NAVAL Group (2020):** Ground robot path optimization. *(Computer vision, YOLO, ROS, Gazebo, SLAM)* .

▶ **Iris recognition & tracking (2018):** Tracking user behavior on mobile devices. *(Computer Vision, YOLO, RestAPI)* .

▶ **Search-based NLP in Otonhanh.vn (2016):** Automobile search platform. *(ML, NLP, Scikit-learn)* .

PUBLICATIONS

“ Pham, V.T., Luu, H.P.H., Le Thi, H.A. (2022). **A Block Coordinate DCA Approach for Large-Scale Kernel SVM**. ICCCI 2022.

“ Pham, V.T., Le Thi, H.A., Luu, H.P.H., Damel, P. (2023). **DCA-Based Weighted Bagging: A New Ensemble Learning Approach**. ACIIDS 2023.

“ Pham, V.T., Le Thi, H.A. and Pascal, D. (2023). **Cost-sensitive weighted bagging DCA based method for imbalanced financial data**. Submitted to NUMTA.

“ Pham, V.T. (2023). **Techniques d'apprentissage automatique basées sur DCA**. Ph.D. Thesis Publication.